

BUSINESS ANALYTICS · PREDICTIVE MODELING

Fitness Market Analysis: LA Fitness Expansion Strategy

Geospatial, Demographic & Regression-Based
Market Potential Assessment — Florida

Exploratory Data Analysis | Regression Results | Predictions & Business Insights
67 Florida Counties · OLS Linear Regression Model

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Exploratory Data Analysis Report

Fitness Market Analysis: LA Fitness Expansion Strategy — Florida

Summary Statistics

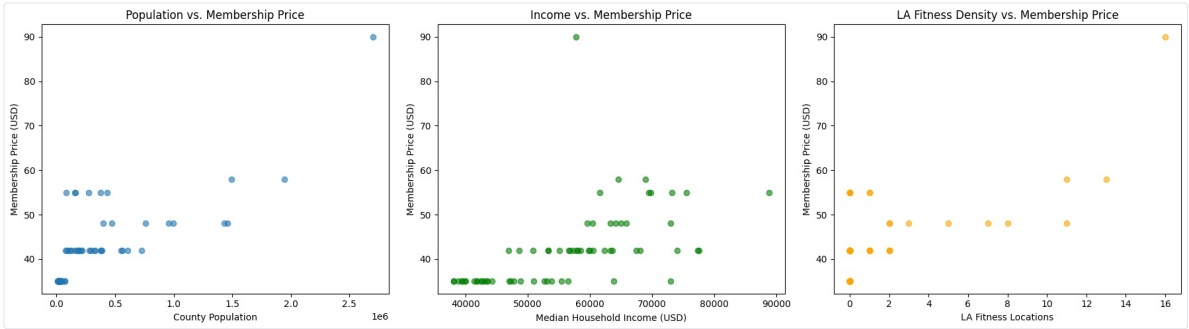
Variable	Mean	Median	Std Dev	Min	Max
MembershipPrice	42.06	42.00	9.09	35.00	90.00
CountyPopulation	321465.48	129752.00	500924.43	7974.00	2701767.00
LAFitnessLocations	1.34	0.00	3.30	0.00	16.00
CompetitorLocations	4.24	2.00	6.70	0.00	29.00
MedianHouseholdIncome	56360.54	56786.00	11801.24	38088.00	88794.00
PopulationDensity	384.49	188.16	574.10	9.54	3504.16
AvgGymRating	4.44	4.44	0.13	4.10	4.80

Correlation Matrix

Variable	MembershipPrice
CountyPopulation	0.785
CompetitorLocations	0.737
LAFitnessLocations	0.707
MedianHouseholdIncome	0.555
PopulationDensity	0.548
AvgGymRating	-0.108

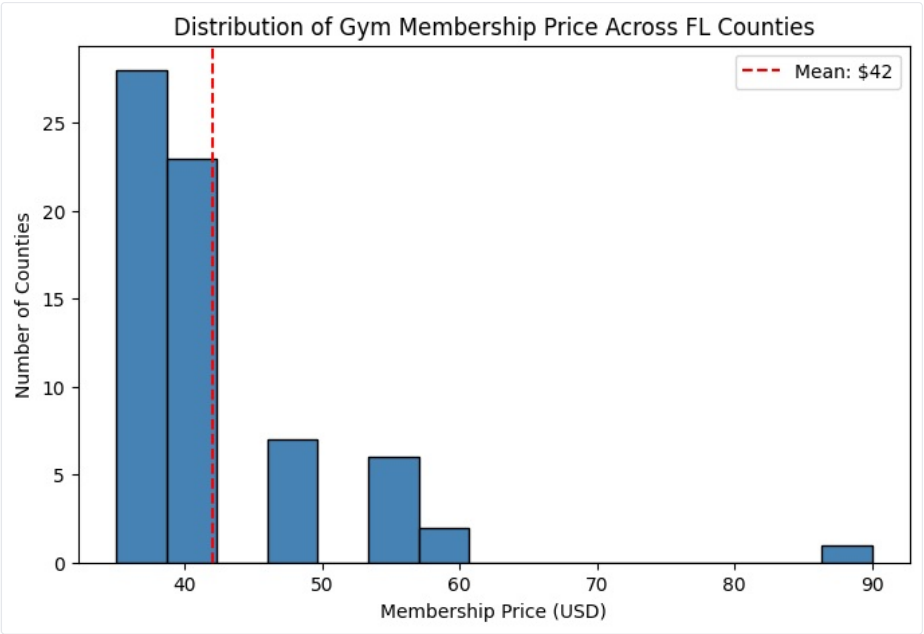
Visualizations

Scatter Plots



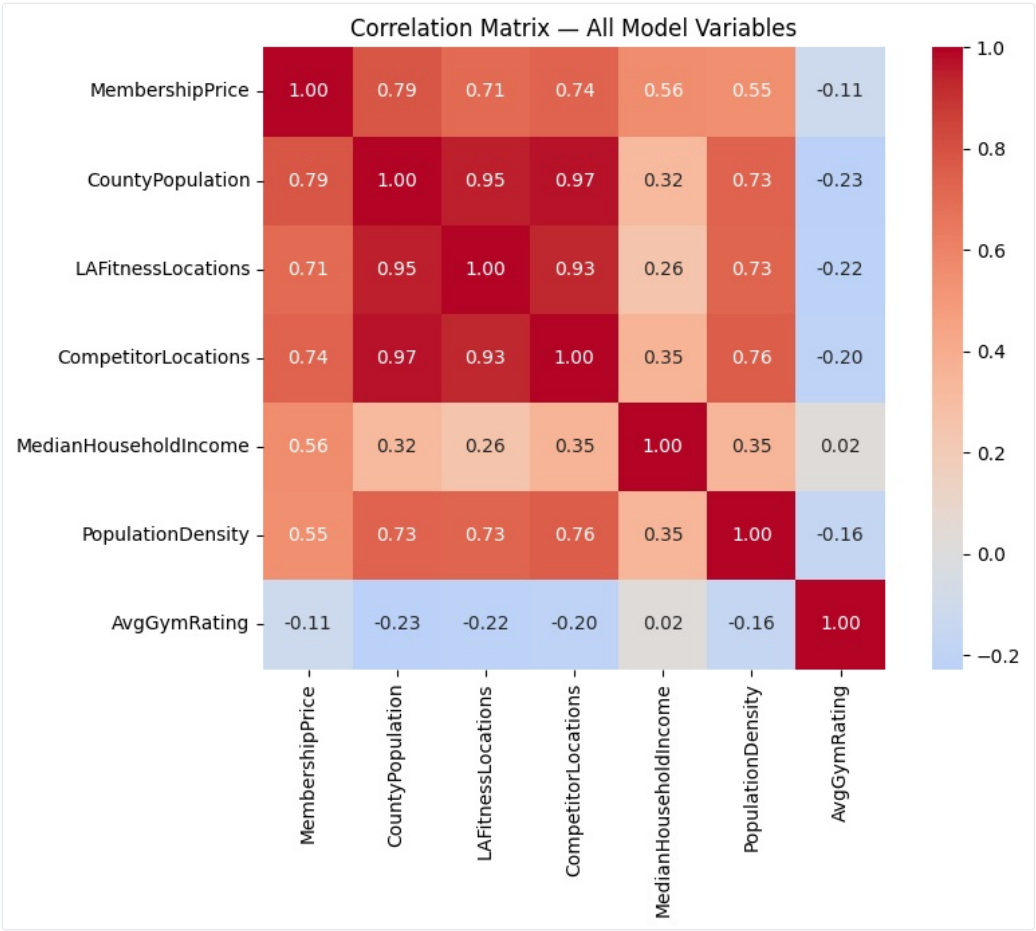
Scatter plots: Population, Income, LA Fitness Locations vs Membership Price

Distribution of Membership Price



Histogram of Membership Price

Correlation Heatmap



Correlation Heatmap

Outlier Detection

IQR bounds: [24.50, 52.50]
Outlier counties (9):

#	CountyName	MembershipPrice	CountyPopulation	LAFitnessLocations
5	Broward	58	1944375	13
10	Collier	55	375752	1
29	Indian River	55	159788	0
41	Martin	55	158431	0
42	Miami-Dade	90	2701767	16
43	Monroe	55	82874	0
49	Palm Beach	58	1492191	11
55	Sarasota	55	434006	1
57	St. Johns	55	273425	0

Business Implications of Outliers

Outliers - Business Implications:

- These nine counties are Florida's premium fitness markets, where above-average membership prices reflect affluent demographics and, in several cases, limited large-chain competition
- Miami-Dade's extreme outlier (\$90) likely stems from high-income urban submarkets supporting boutique and luxury gym pricing
- Notably, six of these counties (Collier, Indian River, Martin, Monroe, Sarasota, St. Johns) have zero or one LA Fitness location despite high prices, suggesting boutique/independent gyms are setting rates in the absence of large-chain competitors.
- In Summary, these underserved, high-income counties could likely support a new LA Fitness location at premium pricing.

Regression Results

OLS Linear Regression — Predicting Florida Gym Membership Price

Full Model — OLS Summary

OLS Regression Results

Dep. Variable:	MembershipPrice	R-squared:	0.754
Model:	OLS	Adj. R-squared:	0.729
Method:	Least Squares	F-statistic:	30.62
Date:	Thu, 09 Jul 2026	Prob (F-statistic):	1.57e-16
Time:	19:23:33	Log-Likelihood:	-195.49
No. Observations:	67	AIC:	405.0
Df Residuals:	60	BIC:	420.4
Df Model:	6		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	6.5922	20.664	0.319	0.751	-34.742	47.927
CountyPopulation	2.475e-05	5.3e-06	4.672	0.000	1.42e-05	3.54e-05
LAFitnessLocations	-0.2002	0.571	-0.351	0.727	-1.342	0.942
CompetitorLocations	-0.7706	0.368	-2.095	0.040	-1.506	-0.035
MedianHouseholdIncome	0.0003	5.49e-05	5.260	0.000	0.000	0.000
PopulationDensity	-0.0015	0.002	-0.926	0.358	-0.005	0.002
AvgGymRating	3.4554	4.646	0.744	0.460	-5.838	12.749

Omnibus:	8.523	Durbin-Watson:	1.711
Prob(Omnibus):	0.014	Jarque-Bera (JB):	9.579
Skew:	0.569	Prob(JB):	0.00832
Kurtosis:	4.462	Cond. No.	2.17e+07

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.17e+07. This might indicate that there are strong multicollinearity or other numerical problems.

Coefficients, p-values, and 95% Confidence Intervals

Variable	Coefficient	Std Error	p-value	CI Lower (95%)	CI Upper (95%)
Intercept	6.5922	20.6642	0.7508	-34.7424	47.9267
CountyPopulation	2.475e-05	5.3e-06	0.000	1.42e-05	3.54e-05
LAFitnessLocations	-0.2002	0.5709	0.7270	-1.3421	0.9417
CompetitorLocations	-0.7706	0.3678	0.0404	-1.5064	-0.0348
MedianHouseholdIncome	0.0003	0.0001	0.0000	0.0002	0.0004
PopulationDensity	-0.0015	0.0016	0.3584	-0.0046	0.0017
AvgGymRating	3.4554	4.6461	0.4600	-5.8382	12.7489

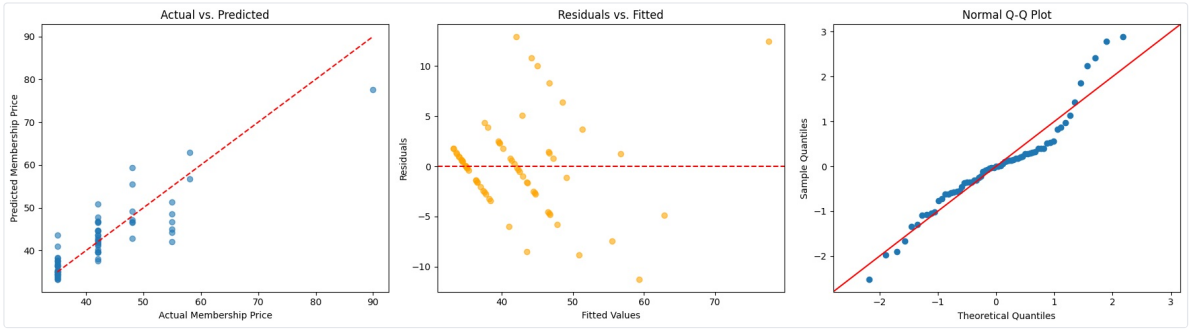
Note: values shown for CountyPopulation (and any variable whose coefficient rounds to 0.0000 at 4 decimal places) are pulled from the full-precision OLS summary above rather than the rounded coefficient table, since the true effect size is non-zero (e.g. ~2.48e-05 per additional resident).

Multicollinearity Check (VIF)

#	Variable	VIF
0	CountyPopulation	20.776945
1	LAFitnessLocations	10.472685
2	CompetitorLocations	17.912376
3	MedianHouseholdIncome	1.238081
4	PopulationDensity	2.431436
5	AvgGymRating	1.068860

VIF > 5 flags a potential multicollinearity concern.

Residual Diagnostics



Actual vs Predicted / Q-Q Plot / Residuals vs Fitted

Normality & Homoscedasticity Tests

Shapiro-Wilk: statistic=0.9394, p-value=0.0027
→ Residuals are NOT normal (p > 0.05 = fail to reject normality)

Breusch-Pagan: LM statistic=27.7835, p-value=0.0001
→ Residuals are heteroscedastic (p > 0.05 = fail to reject homoscedasticity)

Model Refinement (Backward Elimination, p > 0.05 removed)

Dropping 'LAFitnessLocations' (p=0.7270)
Dropping 'AvgGymRating' (p=0.4541)
Dropping 'PopulationDensity' (p=0.3179)

Final predictors: ['CountyPopulation', 'CompetitorLocations', 'MedianHouseholdIncome']

OLS Regression Results

Dep. Variable:	MembershipPrice	R-squared:	0.747
Model:	OLS	Adj. R-squared:	0.735
Method:	Least Squares	F-statistic:	61.99
Date:	Thu, 09 Jul 2026	Prob (F-statistic):	8.84e-19
Time:	19:25:50	Log-Likelihood:	-196.41
No. Observations:	67	AIC:	400.8
Df Residuals:	63	BIC:	409.6
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	21.9303	2.911	7.533	0.000	16.113	27.748
CountyPopulation	2.334e-05	4.56e-06	5.123	0.000	1.42e-05	3.24e-05
CompetitorLocations	-0.8691	0.346	-2.515	0.014	-1.560	-0.178
MedianHouseholdIncome	0.0003	5.26e-05	5.507	0.000	0.000	0.000

Omnibus: 11.402 Durbin-Watson: 1.623
Prob(Omnibus): 0.003 Jarque-Bera (JB): 14.584
Skew: 0.694 Prob(JB): 0.000681
Kurtosis: 4.817 Cond. No. 3.02e+06

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.02e+06. This might indicate that there are strong multicollinearity or other numerical problems.

Full Model vs. Parsimonious Model Comparison

Metric	Full Model	Parsimonious Model
R ²	0.7538	0.7470
Adjusted R ²	0.7292	0.7349
AIC	404.9807	400.8254
BIC	420.4135	409.6442
# Predictors	6.0000	3.0000

Predictions & Business Insights Report

LA Fitness Florida Expansion — Scenario Predictions and Recommendations

Hypothetical Scenarios

Scenario	CountyPopulation	LAFitnessLocations	CompetitorLocations	MedianHouseholdIncome	PopulationDensity	AvgGymRating
A: Rural / Low Competition	50000	2	1	46980.333333	57.796364	4.437581
B: Miami-Dade / High Competition	2600000	15	30	57815.000000	1422.054483	4.340000
C: St. Johns / Untapped Market	273425	1	4	88794.000000	455.219731	4.650000

Predicted Membership Price with 95% Confidence Intervals

Scenario	Predicted Price	CI Lower (avg)	CI Upper (avg)	PI Lower (single county)	PI Upper (single county)
A: Rural / Low Competition	35.82	34.28	37.37	26.34	45.30
B: Miami-Dade / High Competition	73.27	67.25	79.29	62.15	84.39
C: St. Johns / Untapped Market	50.53	46.94	54.12	40.51	60.55

Standardized Coefficients — Relative Influence on Price

Standardized coefficients – relative strength of influence on price:

Variable	Standardized Coefficient
CountyPopulation	1.286
CompetitorLocations	-0.641
MedianHouseholdIncome	0.376

Model Interpretation

Which factors most strongly influence membership pricing?

- Based on the standardized coefficients, county population is the dominant driver of membership price ($\beta = 1.286$), followed by median household income ($\beta = 0.376$), meaning larger and wealthier markets command meaningfully higher prices.
- Competitor locations has a smaller but negative effect ($\beta = -0.641$), showing that pricing power weakens somewhat as more competing gyms enter a market.
- Together these results suggest membership pricing in Florida is set primarily by market size and affluence, with competitive density acting as a moderating rather than dominant force.

How does competition affect price?

- The standardized coefficient for CompetitorLocations is negative (-0.641), indicating that as the number of competing fitness chains in a county rises, average membership price tends to fall
- However, since population and income have larger positive effects, a county can still support high prices despite heavy competition if it's large and affluent enough

Top 5 Expansion Counties (Model-Driven)

#	CountyName	PredictedPrice	CountyPopulation	LAFitnessLocations	CompetitorLocations	ExpansionScore
50	Pasco	45.77	561891	0	7	25717758.70

52	Polk	50.45	725046	1	5	18290104.93
33	Lake	43.91	383956	0	5	16860902.06
40	Marion	41.93	375908	0	4	15762196.68
57	St. Johns	50.53	273425	0	4	13816767.59

Business Recommendations

- Based on the ExpansionScore model, Pasco, Polk, Lake, Marion, and St. Johns counties represent the strongest expansion opportunities, combining sizable populations, predicted membership prices in the 42 to 50 dollars, and little to no existing LA Fitness presence.

Prepared as part of a business analytics project modeling gym membership pricing and expansion potential across Florida counties using Google Places data, U.S. Census demographic data, and an OLS linear regression model.